

PRODUCT REQUIREMENTS DOCUMENT (PRD)

AI Avatar Digital Assistant for QSR Drive-Thru Automation & Personalized Customer Experience

1. Executive Summary

This PRD describes an AI-powered digital avatar designed to enhance the customer experience at QSR drive-thrus. The avatar identifies customers using facial recognition, greets them personally, provides conversational menu guidance, handles ordering in multiple languages, recommends items based on preferences, and processes payments seamlessly using FacePay.

2. Business Objectives

- Reduce customer wait times
- Enhance personalization and loyalty
- Improve order accuracy
- Reduce manual staffing needs
- Increase average order value through tailored recommendations
- Enable seamless, automated payments
- Support global customers via multi-language support

3. Scope

In Scope:

- AI avatar with voice + visual animation
- Facial recognition & customer identification
- Conversational ordering agent
- Menu filtering and intelligent recommendations
- Multi-language support
- FacePay payment authorization
- Integration with POS, KDS, CRM, loyalty systems
- Mobile confirmation notifications
- Privacy & consent workflows

Out of Scope:

- Drive-thru physical redesign
- Human call center services

4. User Roles

- Drive-Thru Customer
- AI Avatar Assistant
- Restaurant Staff
- Store Manager
- Administrator
- Data Analyst
- Payments & Compliance Officer

5. End-to-End Workflow

1. Customer enters QSR lane
2. Avatar activates and scans face
3. Customer is recognized or onboarded
4. Avatar greets customer by name
5. Avatar reviews dietary preferences
6. Menu discussion and filtering
7. AI-driven recommendations
8. Customer finalizes order
9. Avatar requests FacePay consent
10. Payment completed via facial authentication
11. Confirmation sent to mobile
12. Order sent to kitchen
13. Customer picks up order

6. Functional Requirements

6.1 Facial Recognition & Identification

- Detect and identify customers under varied lighting
- Retrieve saved preferences and loyalty data
- Support new customer enrollment
- Provide fallback when recognition fails

6.2 AI Avatar Behavior & Interface

- Lifelike 3D avatar with expressions
- Lip-sync accuracy
- Brand-aligned personality
- Seasonal visual themes

6.3 Conversational Ordering Assistant

- Real-time, natural voice interaction
- Accent and noise-resistant recognition
- Handle menu questions, customization and modifiers
- Multi-turn conversations

6.4 Multi-Language Support

- Automatic language detection
- Region-specific language support
- Accurate translation of menu items
- Allow switching languages mid-session

6.5 Personalization Engine

- Retrieve dietary preferences and restrictions
- Identify favorite and frequently ordered items
- Provide upsell and promotion suggestions
- Filter menu based on stored preferences

6.6 Menu Logic & Recommendations

- Real-time menu availability
- Combo suggestions
- Exclude unavailable or dietary-restricted items
- Prioritize profitable or trending items

6.7 Order Capture

- Add, remove, and edit items
- Understand special requests (e.g., no onions)
- Provide full order confirmation
- Undo or restart options

6.8 FacePay Payment Processing

- Request explicit consent
- Authenticate identity via facial scan
- Auto-charge linked card or wallet
- Send payment receipt to mobile app
- Provide fallback methods if needed

6.9 System Integrations

- POS and KDS
- Loyalty and CRM systems
- Payment gateways
- Customer mobile app

6.10 Privacy & Compliance

- Consent-based facial data usage
- Clear opt-in and opt-out mechanisms
- Biometric data encryption
- Compliance with GDPR, CCPA, etc.
- Audit trails

6.11 Failover Handling

- Graceful fallback to human operator
- Auto-detection of camera or audio issues
- Clear user messaging

7. Non-Functional Requirements

- High availability (99.9%)
- Low response latency (<1 sec)
- Scalable for large QSR chains
- Background noise resistance
- Secure biometric and payment processing
- Extensible for future capabilities

8. Use Cases

UC1: Returning Customer Personalized Ordering

Avatar identifies customer → greets by name → recalls usual items and preferences → offers personalized suggestions

UC2: New Customer Profile Creation

Avatar requests permission → scans face → captures preferences → begins ordering interaction

UC3: Multi-Language Ordering

Customer speaks preferred language → avatar responds accordingly → orders in selected language

UC4: Dietary-Filtered Menu View

Customer is vegetarian → avatar filters menu → presents veg-only options

UC5: Intelligent Upselling

Based on user history → avatar suggests suitable add-ons and offers

UC6: FacePay Payment

Avatar requests consent → facial authentication → payment auto-processed → receipt sent

UC7: System Failover

Equipment malfunction detected → user routed to human operator

9. Futuristic Use Cases

- Emotion-aware chatbot adapting tone dynamically
- Gesture-based confirmation (nod = yes)
- Predictive ordering before customer arrives
- Avatar personalization selected by user
- Integration with in-car infotainment systems
- Health-conscious menu suggestions based on user profile

10. Success Metrics

- Decrease in average service time per car
- Increased drive-thru throughput
- Increased AOV via targeted upselling
- Reduced manual staffing needs
- Higher customer satisfaction and retention
- Improved order accuracy